

How to run and use the demo

Everything you need to open the demo, use each tab with your own Marathi text, and switch on the full models. No technical background needed.

QUICK START – 30 SECONDS

1. Open the file `Marathi-NLP-Demo.html` (double-click it).
2. Click a tab: **Morphology**, **Search** or **Full pipeline**.
3. Type or paste Marathi text. Results appear on the spot.

What's in this manual

1. What you need
2. Opening the demo
3. The screen at a glance
4. Tab 1 — Morphology
5. Tab 2 — Search
6. Tab 3 — Full pipeline
7. English translation
8. Using the full models (optional)
9. The Streamlit dashboard (optional)
10. Problems and fixes
11. Checklist before presenting

1 What you need

The demo can run in three ways. Most of the time you only need the first one.

WAY TO RUN	WHAT YOU NEED	WHAT YOU GET
A. Just the web page <i>recommended</i>	Any browser (Chrome, Edge, Firefox, Safari). No installation.	All three tabs work on any text you paste. Category, sentiment and entities use small built-in models. Translation needs internet.
B. Web page + model server	Python 3.10 or newer, plus one command left running.	Same page, but category, entities, sentiment and translation use the full, more accurate models .
C. Streamlit dashboard	Python, and about 10 minutes of one-time setup.	A second app that also searches all 3,000 articles and can fetch today's news.

2 Opening the demo

If you already have the project folder

- 1 Open the folder `Marathi-news-intelligence-pipeline`.
- 2 Double-click `Marathi-NLP-Demo.html`. It opens in your web browser.

If you are getting it from GitHub

The project is at `github.com/anishc23/nlp-project`. GitHub shows HTML files as code, so you have to download the page first:

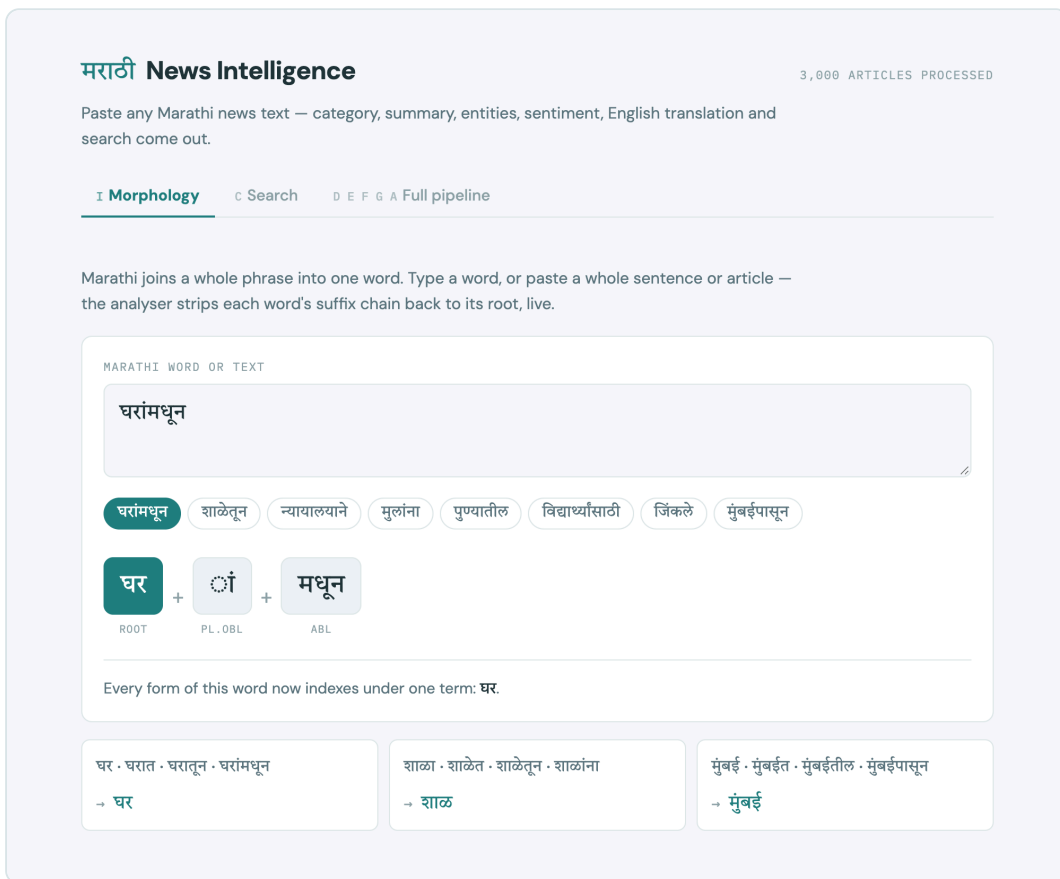
- 1 Go to the repository and open `Marathi-news-intelligence-pipeline`, then `Marathi-NLP-Demo.html`.
- 2 Click the **Download raw file** button (the arrow icon at the top right of the file view).
- 3 Double-click the downloaded file.

To get the whole project instead, click the green **Code** button, then **Download ZIP**, and unzip it.

GOOD TO KNOW

The page is one file of about 4 MB. You can email it, copy it to a pendrive, or open it on a phone. Everything except translation works offline. The internet is only used for nicer fonts and for English translation.

3 The screen at a glance



What you see when the page opens. The Morphology tab is already selected.

There are three tabs across the top. The small letters before each name are the NLP task codes from the assignment list:

TAB	TASK CODES	WHAT IT DOES
Morphology	I	Breaks Marathi words into a root plus grammar endings.
Search	C	Searches 400 news articles, with and without the root-finding step, side by side.

4 Tab 1 – Morphology

Marathi joins what English says in several words into one word. For example घरामधून means "from inside the houses". This tab splits such words back into their pieces.

Look at one word

- 1 Type a Marathi word in the box, or click one of the grey example buttons under it.
- 2 The coloured blocks show the result. The dark green block is the **root** (the basic word). The grey blocks are the **endings** that were removed.

MARATHI WORD OR TEXT

विद्यार्थ्यांसाठी

घरामधून

शाळेतून

न्यायालयाने

मुलांना

पुण्यातील

विद्यार्थ्यांसाठी

जिकले

मुंबईपासून

विद्यार्थ्य

+

ां

+

साठी

ROOT

PL. OBL

BEN

Every form of this word now indexes under one term: विद्यार्थ्य.

विद्यार्थ्यांसाठी ("for the students") = root + plural ending + "for"

Paste a sentence or a whole article

- 1 Paste any Marathi text into the same box.
- 2 You get a table with every word, its root, the endings removed and their grammar labels, plus three numbers at the top: how many words, how many had an ending, and how many different words are left after reducing them to roots.

मुंबईतील शाळांमध्ये विद्यार्थ्यांसाठी नवीन अभ्यासक्रम सुरू झाला. सरकारने याबाबत घोषणा केली.

घरांमधून

शाळेतून

न्यायालयाने

मुलांना

पुण्यातील

विद्यार्थ्यांसाठी

जिकले

मुंबईपासून

11 27.3% 11 → 11

words

carry a suffix

distinct forms → roots

WORD	ROOT	SUFFIXES STRIPPED	GRAMMAR
मुंबईतील	मुंबई	exception list	
शाळांमध्ये	शाळ	ां + मध्ये	PL.OBL + LOC
विद्यार्थ्यांसाठी	विद्यार्थ्य	ां + साठी	PL.OBL + BEN
नवीन	नवीन	—	
अभ्यासक्रम	अभ्यासक्रम	—	
सुरू	सुर	final vowel normalised	
झाला	झाल	final vowel normalised	
सरकारने	सरकार	exception list	
याबाबत	याबाब	त	LOC
घोषणा	घोषण	final vowel normalised	

A pasted sentence. Every word is reduced to its root.

What the notes in the table mean

—

The word had no ending. It is already a root.

EXCEPTION LIST

A word the analyser handles with a fixed answer, so it is not cut wrongly (for example सरकारने stays सरकार).

FINAL VOWEL
NORMALISED

Only the last vowel sign was dropped, so related forms (शाळा, शाळेत) end up with the same root.

Common grammar labels

LABEL	MEANING	EXAMPLE ENDING
PL.OBL	plural form before another ending	ां
LOC	"in / at"	त, मध्ये
ABL	"from"	पासून, मधून, तून
GEN	"of"	चा, ची, चे, च्या
DAT	"to / for (a person)"	ला, ना
INS	"by"	ने, नी

BEN	"for (the sake of)"	साठी
PST / FUT / PRS	past / future / present verb endings	ले, णार, तो

The three cards at the bottom of the tab always show four forms of one word collapsing to a single root:

घर • घरात • घरातून • घरांमधून → घर	शाळा • शाळेत • शाळेतून • शाळांना → शाळ	मुंबई • मुंबईत • मुंबईतील • मुंबईपासून → मुंबई
---------------------------------------	---	---

Four forms, one root. This is what makes the search tab work.

5 Tab 2 — Search

There are two search engines over the same 400 news articles. The only difference: the right-hand one reduces every word to its root first (using the Morphology tab's analyser); the left-hand one does not.

Run a search

- 1 Click the **Search** tab. The query न्यायालय ("court") is already filled in.
- 2 Type or paste any Marathi word or phrase in the box, or click a grey example button.
- 3 Compare the two big numbers: results **without** morphology (left, red) and **with** it (right, green).

SEARCH QUERY

न्यायालय

न्यायालय

सरकार

महिला

शाळा

पोलीस

पाऊस

WITHOUT MORPHOLOGY

1

कोपडी निकाल : कोर्टूममधली ती 8 मिनिट!

CRIME · SCORE 5.80

WITH MORPHOLOGY

14

कोपडी निकाल : कोर्टूममधली ती 8 मिनिट!

CRIME · SCORE 5.64

भारतीय न्यायव्यवस्थेसमोर अभूतपूर्व संकट : सरन्यायाधीशच संशयाच्या भोवऱ्यात

EDUCATION · SCORE 4.92

फी वाढ : 'शाळा बंद मग पूर्ण फी का द्यायची?' पालकांचा सवाल; काय आहे यासंबंधीचा नियम?

EDUCATION · SCORE 4.86

बाबरी मशीद प्रकरणात आतापर्यंत काय काय झालं?

MANORANJAN · SCORE 3.75

Morphology found 13 more articles the plain index cannot reach.

► Add your own articles to the index

न्यायालय: 1 result without morphology, 14 with it.

The line under the results explains what happened. For some words both sides find the same number, because the articles use the word in exactly the same form:

SEARCH QUERY

पाऊस

न्यायालय

सरकार

महिला

शाळा

पोलीस

पाऊस

WITHOUT MORPHOLOGY3

कृत्रिम पाऊस: 30 कोटींची तरतूद पण यावेळेस तरी पाऊस पडेल का?
INTERNATIONAL · SCORE 7.39

निसर्ग चक्रीवादळ: दिशा बदलून पुण्यालाही तडाखा; कोकणातल्या नुकसानीचा आढावा सुरू
TRAVEL · SCORE 5.16

लेगॉस, नायजेरिया: 'या' देशात तरंगतं शहर का तयार केलं जात आहे?
INTERNATIONAL · SCORE 4.05

WITH MORPHOLOGY3

कृत्रिम पाऊस: 30 कोटींची तरतूद पण यावेळेस तरी पाऊस पडेल का?
INTERNATIONAL · SCORE 7.41

निसर्ग चक्रीवादळ: दिशा बदलून पुण्यालाही तडाखा; कोकणातल्या नुकसानीचा आढावा सुरू
TRAVEL · SCORE 5.15

लेगॉस, नायजेरिया: 'या' देशात तरंगतं शहर का तयार केलं जात आहे?
INTERNATIONAL · SCORE 4.07

No difference for this query — the articles happen to use it in the same form.

► Add your own articles to the index

पाऊस ("rain"): 3 and 3, no difference.

Add your own articles to the search

- 1 Under the results, click **Add your own articles to the index**.
- 2 Paste a headline and the article text, then click **Add to both indexes**.
- 3 Search for a word from your article. It appears in the results marked **YOUR ARTICLE**, with a topic chosen automatically.

▼ Add your own articles to the index

HEADLINE

पुण्यात नवीन मेट्रो मार्ग

MARATHI ARTICLE TEXT

पुणे महानगरपालिकेने शहरात नवीन मेट्रो मार्गाची घोषणा केली. या मेट्रोमुळे पुणेकरांचा प्रवास सोपा होणार आहे.

Add to both indexes

Pasting an article

SEARCH QUERY

मेट्रो

न्यायालय

सरकार

महिला

शाळा

पोलीस

पाऊस

WITHOUT MORPHOLOGY1

पुण्यात नवीन मेट्रो मार्ग
YOUR ARTICLE · POLITICS · SCORE 10.31

WITH MORPHOLOGY1

पुण्यात नवीन मेट्रो मार्ग
YOUR ARTICLE · POLITICS · SCORE 11.29

No difference for this query — the articles happen to use it in the same form.

▼ Add your own articles to the index

HEADLINE

MARATHI ARTICLE TEXT

Add to both indexes

Added as Politics. Your articles in the index: 1.

Finding it again

NOTE

Added articles are kept only until you reload the page.

6 Tab 3 — Full pipeline

This tab takes a whole news article and runs every stage on it.

Analyse an article

- 1 Click the **Full pipeline** tab. The first example is already analysed.
- 2 To try your own: click **Clear**, paste a headline (optional) and the Marathi article text. A copied news article from any Marathi site works well.
- 3 Click **Analyse** (or press **Ctrl + Enter**, **⌘ + Enter** on a Mac).
- 4 Or pick one of the eight ready-made examples from **Start from an example**.

START FROM AN EXAMPLE

— paste your own below —

HEADLINE (OPTIONAL)

भारताचा इंग्लंडवर शानदार विजय

MARATHI ARTICLE TEXT

भारतीय क्रिकेट संघाने दुसऱ्या कसोटी सामन्यात इंग्लंडचा नऊ गडी राखून पराभव केला. कर्णधार रोहित शर्माने मुंबईत पत्रकार परिषदेत संघाचे कौतुक केले. मात्र दुखापतीमुळे जसप्रीत बुमराह पुढील सामन्याला मुकणार आहे. हा सामना वानखेडे स्टेडियमवर खेळवण्यात आला होता. प्रेक्षकांनी मोठ्या संख्येने उपस्थित राहून संघाला पाठिंबा दिला.

Analyse Clear IN-BROWSER MODELS

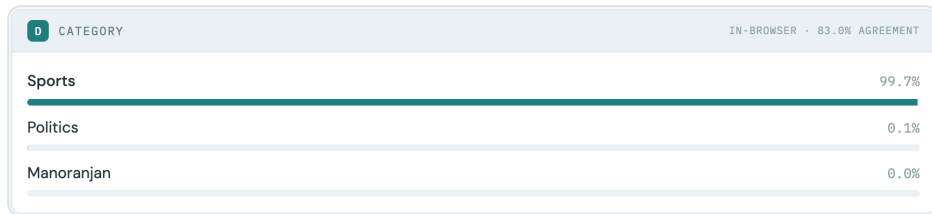
A pasted article, ready to analyse. The badge shows which models are being used.

Reading the results

The results appear below as a stack of boxes, one per stage, in the order the pipeline runs them.

BOX	WHAT IT SHOWS
Input article	Your headline, how many words it has, and what share of them carry a grammar ending.
D · Category	The three most likely topics out of 12 (Politics, Sports, Health...) with confidence bars.

E • Summary	The three most important sentences of the article, in their original order.
G • Entities	Names found in the text, grouped as Person, Organization, Location, Date, Designation and so on.
F • Sentiment	The overall tone (Positive, Neutral or Negative), a colour bar, and the tone of each summary sentence.
A • English translation	A button that translates the headline and summary to English (see section 7).



Category box



Sentiment box, with the tone of each sentence

The small grey label on each box

The top-right corner of each box says which model produced it:

IN-BROWSER • 83% AGREEMENT	A small model built into the page. The number is how often it gives the same answer as the full model, measured on 1,362 articles it had never seen.
FULL MODEL	The real, large model, used when the model server is running (section 8).
TEXTTRANK • SAME ALGORITHM AS PYTHON	The summary uses exactly the same method as the Python project, so it is identical either way.

EXPECT SOME MISTAKES FROM THE IN-BROWSER MODELS

The built-in entity finder is the least accurate part (it agrees with the full model about 72% of the time). In the example above it wrongly marks विजय ("victory") as a person's name. For the most accurate results, start the model server (section 8).

7 English translation

- 1 After analysing, scroll to the A • English translation box and click Translate to English.

- 2 The first time only, the page downloads the translation model (about 900 MB). A progress line shows how much has arrived. This takes a few minutes on a normal connection.
- 3 The headline and summary appear in English. The model is saved in the browser, so later translations skip the download and take only seconds. Keep an internet connection on, though: the page still loads the small translation program itself from the internet.

A

ENGLISH TRANSLATION

NLLB-200 · LOCAL SERVER

India's glorious victory over England

The Indian cricket team defeated England by nine wickets in the second Test. However, due to injury, Jaspreet Bumrah will miss the next match. The match was played at the Wankhede Stadium.

NLLB-200 on the local server.

The translated headline and summary

TIPS

- For the ready-made examples there is also a **Show saved translation** button, which is instant.
- Do the first translation at home, before the presentation, so the model is already saved.
- If the model server is running, translation goes through it instead: no download, and faster.

8 Using the full models (optional)

The page can hand your text to the project's real, full-size models, which are more accurate. You need Python 3.10 or newer.

One-time setup

Open a terminal (Mac: **Terminal**; Windows: **Command Prompt**) in the `Marathi-news-intelligence-pipeline` folder and run:

```
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

Every time

- 1 In that folder, run `python app/server.py` and leave the window open.
- 2 Open (or reload) `Marathi-NLP-Demo.html`.
- 3 Click **Analyse**. The badge next to the button changes to **Full models · local server**, and the boxes say **full model**.
- 4 When finished, go back to the terminal and press **Ctrl + C** to stop the server.

Analyse

Clear

● IN-BROWSER MODELS

Analyse

Clear

● FULL MODELS · LOCAL SERVER

Without the server

With the server running

The difference shows most in the entities. The same article, with the full model:

ENTITIES

FULL MODEL

PERSON

जसप्रीत बुमराह

LOCATION

इंग्लंडवर

भारताचा

वानखेडे स्टेडियमवर

MEASURE

दुसऱ्या

नऊ

Full model: correctly finds जसप्रीत बुमराह as the only person.

GOOD TO KNOW

- The **first** Analyse after starting the server downloads the models (about 5 GB, once) and loads them. That first run can take a while; after that each article takes a few seconds.
- The server only accepts connections from your own computer.
- If the server stops, the page automatically goes back to its built-in models.

9 The Streamlit dashboard (optional)

A second app that works over all 3,000 processed articles. After the one-time setup in section 8, run these once (they take about 10 minutes and download the datasets):

```
python scripts/download_data.py
python scripts/build_corpus.py --limit 3000
python scripts/build_index.py --plain-bm25
```

Then start it any time with:

```
streamlit run app/dashboard.py
```

It opens in the browser with five tabs: **Search** (all 3,000 articles, with topic and tone filters), **Overview** (topic and tone charts), **Entity tone** (how a person or party is covered), **Morphology**, and **Analyse** (paste text, or fetch today's news from BBC Marathi, ABP Majha and Lokmat).

10 Problems and fixes

PROBLEM	WHAT TO DO
GitHub shows lines of code instead of the page	Use Download raw file and open the downloaded file (section 2).

Marathi letters look plain or slightly different	That is only the font; without internet the page uses your computer's Marathi font. Everything still works.
"That doesn't look like Marathi text"	The page only analyses text in Devanagari script. Paste Marathi, not English or romanised Marathi.
The summary says "Too short to summarise"	No sentence had three or more words. Paste full sentences ending with ! . ! or ?
The summary is the whole text	With three sentences or fewer there is nothing to shorten. Paste a longer article.
"Translation failed"	Translation needs an internet connection (unless the model server is running). Connect and click the button again.
Translation is very slow the first time	It is downloading about 900 MB. Watch the progress line; it only happens once per browser.
Badge still says "In-browser models" with the server running	Check the terminal still shows the server, then click Analyse again (the page checks on every click). The server must use port 8765; close any other program on that port.
My added search articles disappeared	They are kept only until the page is reloaded. Add them again.
The browser's developer console shows "connection refused"	Harmless. It is the page checking whether the model server is running.

11 Checklist before presenting

- 1 Open the demo page on the presentation laptop and click through all three tabs once.
- 2 With internet on, click **Translate to English** once so the translation model is downloaded and saved.
- 3 Copy a short Marathi news article (4–8 sentences) from a news site, ready to paste into the Full pipeline tab.
- 4 Optional: start `python app/server.py` and run **Analyse** once, so the full models are already loaded.
- 5 Keep this manual and the Explanation Guide open in another window.